3679. Minimum Discards to Balance Inventory

Solved

Medium Topics Hint

You are given two integers `w` and `m`, and an integer array `arrivals`, where `arrivals[i]` is the type of item arriving on day `i` (days are **1-indexed**).

Items are managed according to the following rules:

- Each arrival may be **kept** or **discarded**; an item may only be discarded on its arrival day.
- For each day `i`, consider the window of days `[max(1, i - w + 1), i]` (the `w` most recent days up to day `i`):
  - For **any** such window, each item type may appear **at most** `m` times among kept arrivals whose arrival day lies in that window.
  - If keeping the arrival on day `i` would cause its type to appear **more than** `m` times in the window, that arrival **must** be discarded.

Return the **minimum** number of arrivals to be discarded so that every `w`-day window contains at most `m` occurrences of each type.

Example 1:

> **Input:** arrivals = [1,2,1,3,1], w = 4, m = 2
>
> **Output:** 0
>
> **Explanation:**
>
> - On day 1, Item 1 arrives; the window contains no more than `m` occurrences of this type, so we keep it.
> - On day 2, Item 2 arrives; the window of days 1 - 2 is fine.
> - On day 3, Item 1 arrives, window `[1, 2, 1]` has item 1 twice, within limit.
> - On day 4, Item 3 arrives, window `[1, 2, 1, 3]` has item 1 twice, allowed.
> - On day 5, Item 1 arrives, window `[2, 1, 3, 1]` has item 1 twice, still valid.
>
> There are no discarded items, so return 0.

Example 2:

> **Input:** arrivals = [1,2,3,3,3,4], w = 3, m = 2
>
> **Output:** 1
>
> **Explanation:**
>
> - On day 1, Item 1 arrives. We keep it.
> - On day 2, Item 2 arrives, window `[1, 2]` is fine.
> - On day 3, Item 3 arrives, window `[1, 2, 3]` has item 3 once.
> - On day 4, Item 3 arrives, window `[2, 3, 3]` has item 3 twice, allowed.
> - On day 5, Item 3 arrives, window `[3, 3, 3]` has item 3 three times, exceeds limit, so the arrival must be discarded.
> - On day 6, Item 4 arrives, window `[3, 4]` is fine.
>
> Item 3 on day 5 is discarded, and this is the minimum number of arrivals to discard, so return 1.

Constraints:

- `1 <= arrivals.length <= 10^5`
- `1 <= arrivals[i] <= 10^5`
- `1 <= w <= arrivals.length`
- `1 <= m <= w`

Seen this question in a real interview before? 1/6

Yes No

Accepted 20,077 /57.3K | Acceptance Rate 35.0 %

Topics

Hint 1

Hint 2

Hint 3

Hint 4

Discussion (42)

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